Analysis of the CoAP data within a DDOS attack

Project Presentation

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Contents

Intro

Motivation

- starting at the 25th of August 2025 there was a DDOS attack on mobi3
- part of that multi-protocol DDOS attack was CoAP traffic
- CoAP is also the focus of my work at RIOT and in other modules

Amplifications Attacks using CoAP

- based upon the IRTF draft: *Amplification Attacks Using the Constrained Application Protocol (CoAP)* [1]
- are a type of reflection-based volumetric DDoS/DoS attacks
- TODO: which amplification factor, see [1]

Amplifications Attacks using CoAP

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- Simple Amplification Attacks
- Amplification Attacks using the Observe Option
- Amplification Attacks using Group Requests
- Amplification Attacks using Group Requests and the Observe Option

Research Question

Are the 4 types of amplification attacks recognizable in the DDOS attack on mobi3?

Method

- 779 GiB of data: *net/archive/ddos_data*
- only a subset of data was used (next slide)
- parsed with *tshark*
- *general_data_set* and *coap_data_set*

Method

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```
All pakets           :  696,340,178
All bytes            : 801,727,874,184  (801727.9 MB)
Timepoints           : 2025-08-13 15:51:49.573544  →  2025-08-13 17:46:30.464776
Duration             : 6880.9s / 114.7min / 1.9hours
Ø Packetrate         :  101,199.1 pkt/s
Ø Bitrate            :      932.1 Mbit/s
Average packet size  : 1151.3 bytes
Maximum packet size  : 18890 bytes
```

Figure 1: _general_data_set_

Method

HAW HAMBURG

```
CoAP distribution   : 0.07%
All pakets          :  696,340,178
CoAP packets        :   51,418,885
All bytes           : 801,727,874,184  (801727.9 MB)
CoAP bytes          : 68,581,946,168  (68581.9 MB)
Ø CoAP Packetrate   :  101,199.1 pkt/s
Ø CoAP Bitrate      :      932.1 Mbit/s
Average packet size : 1333.8 bytes
Maximum packet size : 1514 bytes
```

Figure 2: _coap_data_set_

Plots

General Analysis

```
All pakets          :  696,340,178
All bytes           : 801,727,874,184  (801727.9 MB)
Timepoints          : 2025-08-13 15:51:49.573544  →  2025-08-13 17:46:30.464776
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Average packet size : 1151.3 bytes
Maximum packet size : 18890 bytes
```

Figure 3: Overview of analyzed data

General Analysis

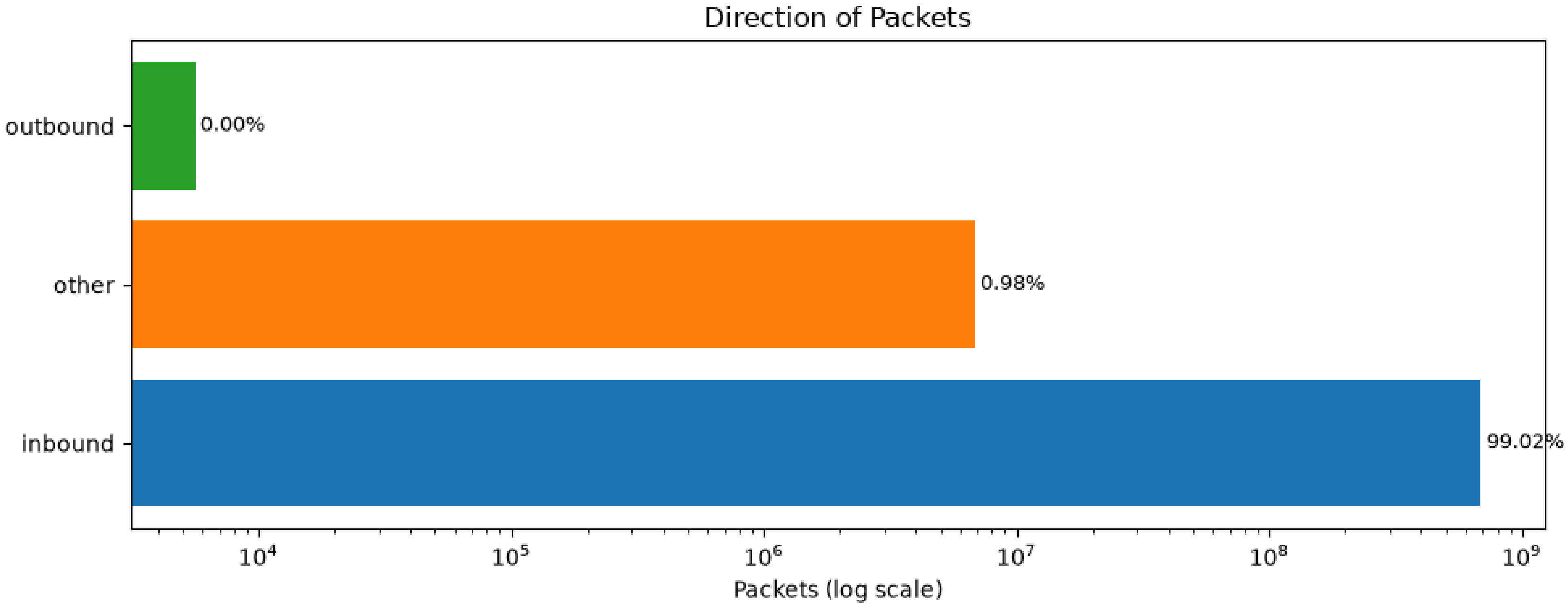

Figure 4: Share of Packet Directions

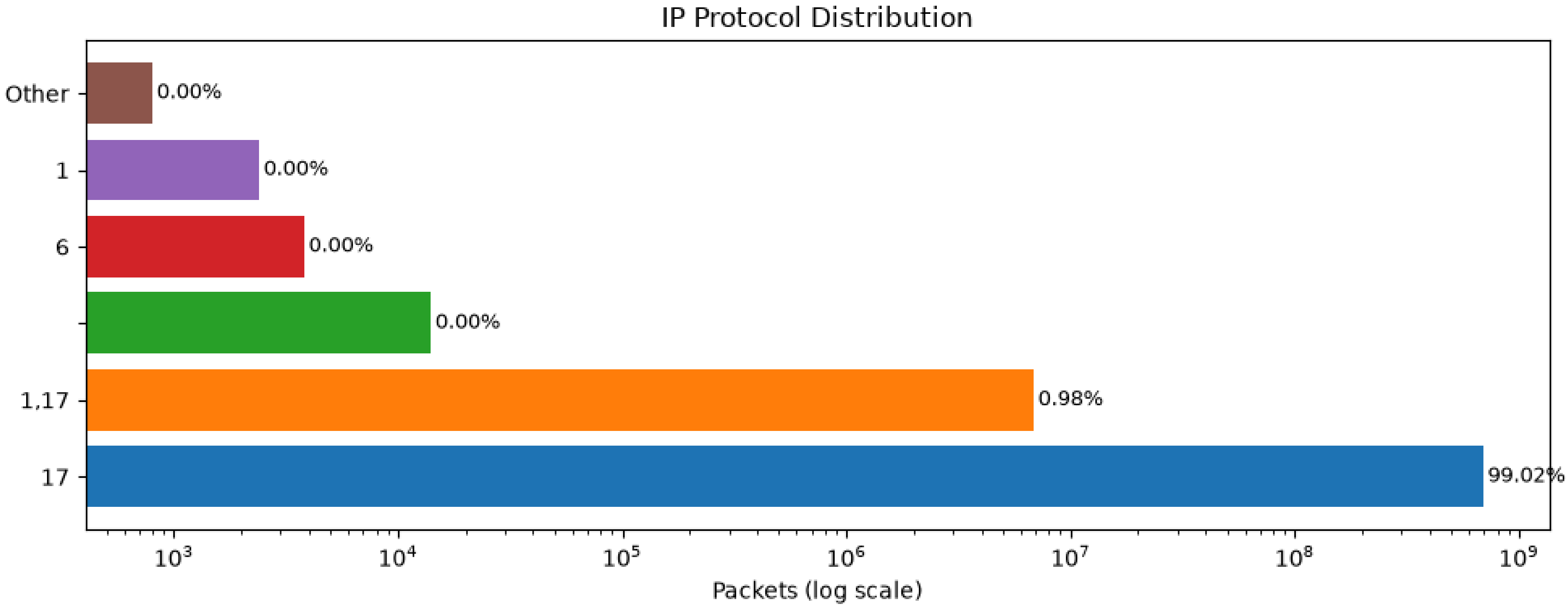

Figure 5: Share of IP Protocols

General Analysis

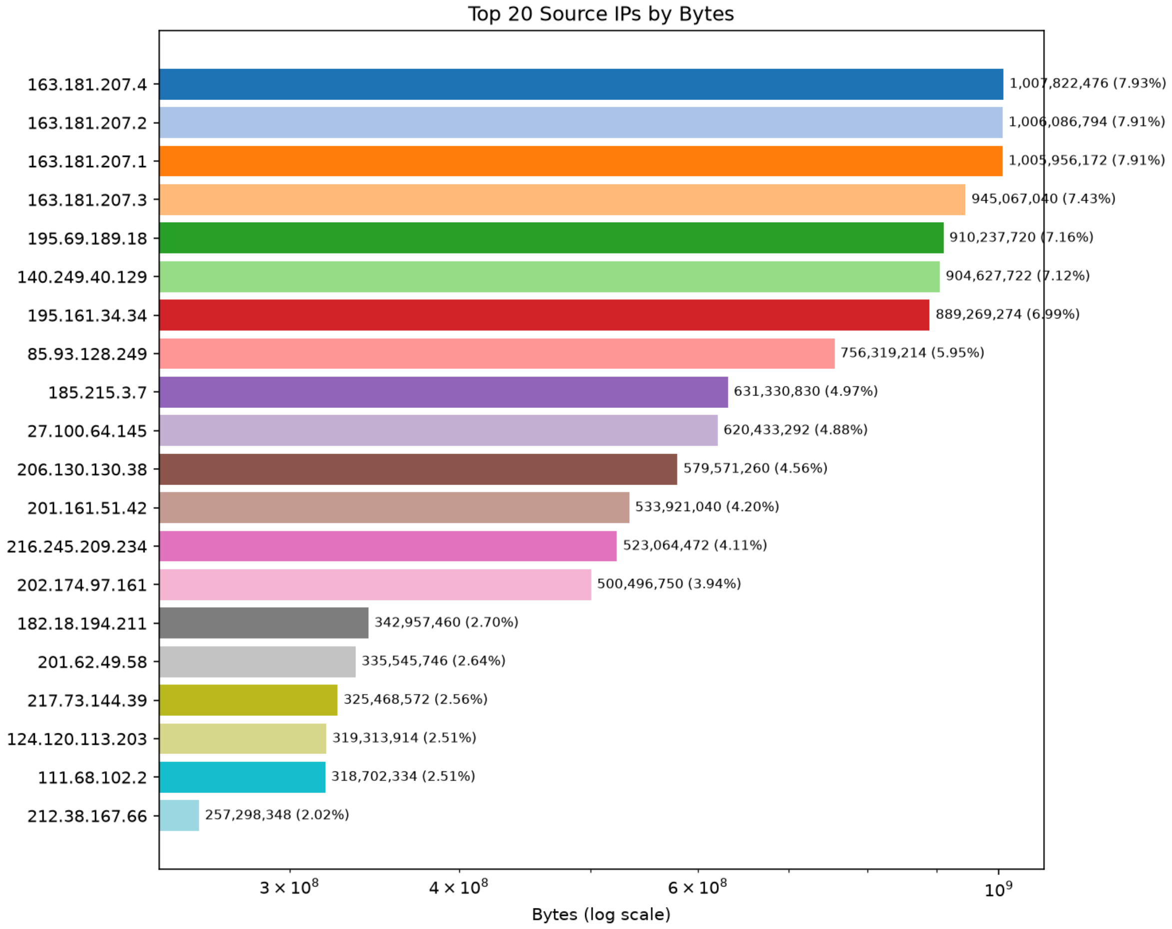

Figure 6: Top Talkers

CoAP specific Analysis


```
CoAP distribution   : 0.07%
All pakets          :  696,340,178
CoAP packets        :   51,418,885
All bytes           : 801,727,874,184  (801727.9 MB)
CoAP bytes          : 68,581,946,168  (68581.9 MB)
Ø CoAP Packetrate   :  101,199.1 pkt/s
Ø CoAP Bitrate      :      932.1 Mbit/s
Average packet size : 1333.8 bytes
Maximum packet size : 1514 bytes
```

Figure 7: Overview of CoAP data

CoAP specific Analysis

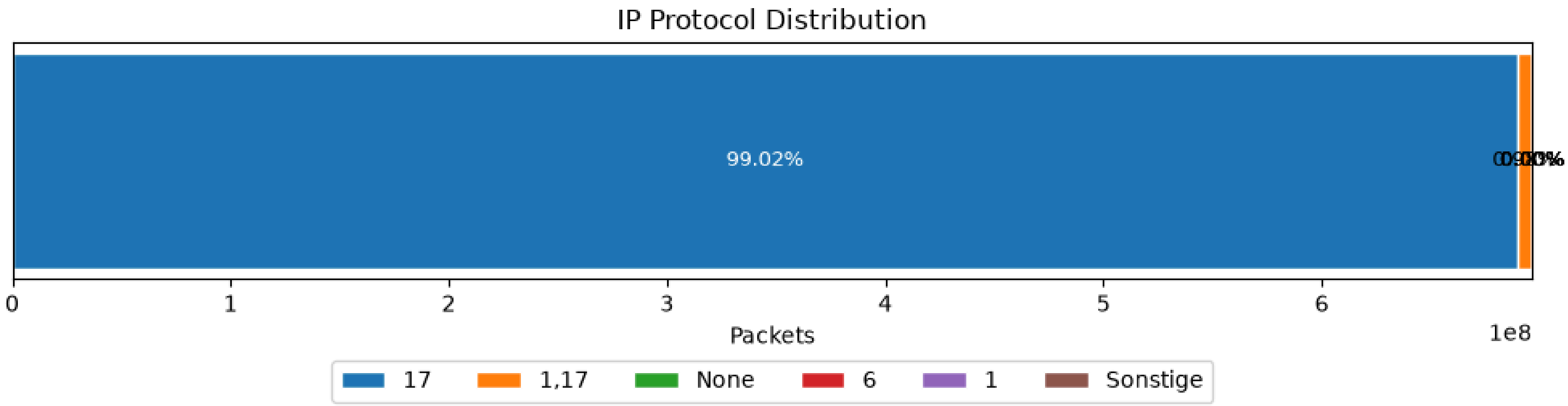

Figure 8: Share of IP Protocols of CoAP packets

CoAP specific Analysis

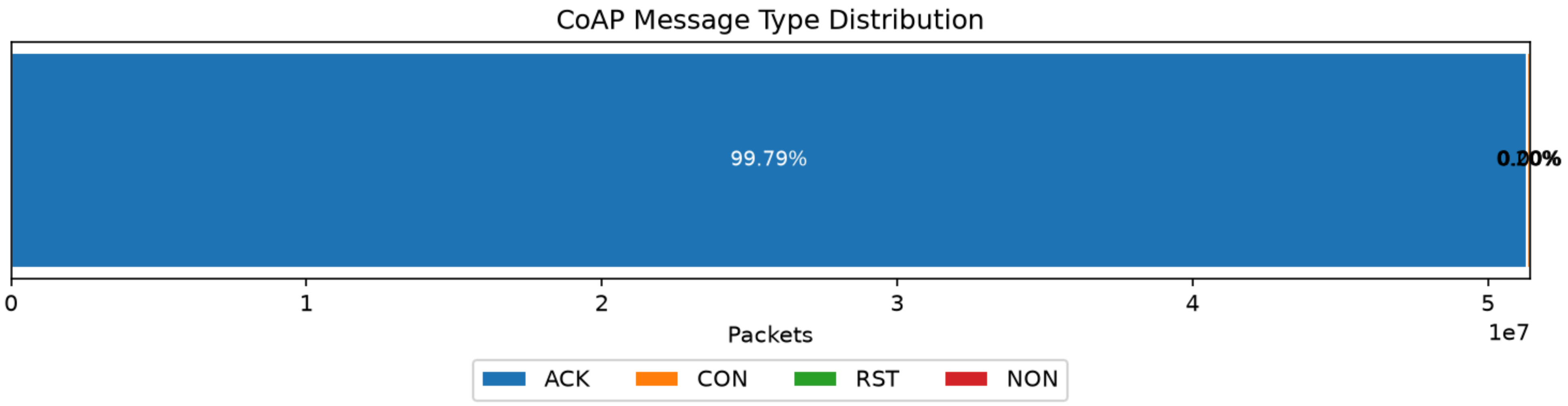

Figure 9: Share of CoAP Message Types

CoAP specific Analysis

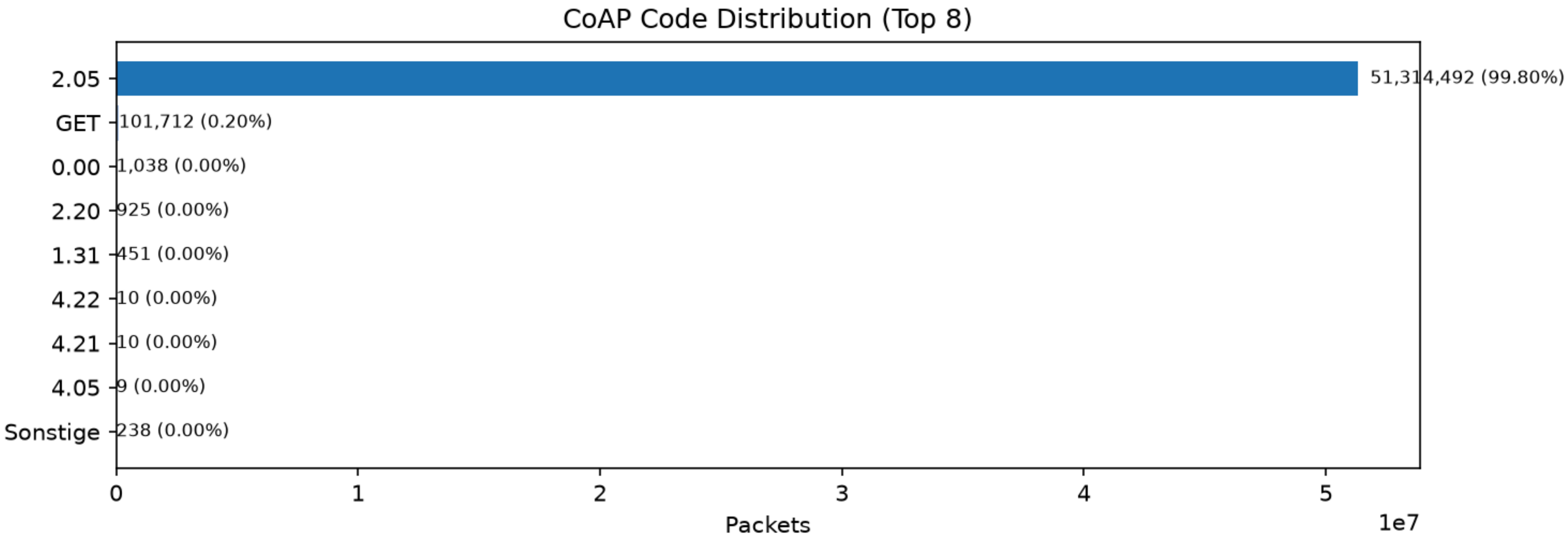

Figure 10: Share of CoAP Codes

CoAP specific Analysis

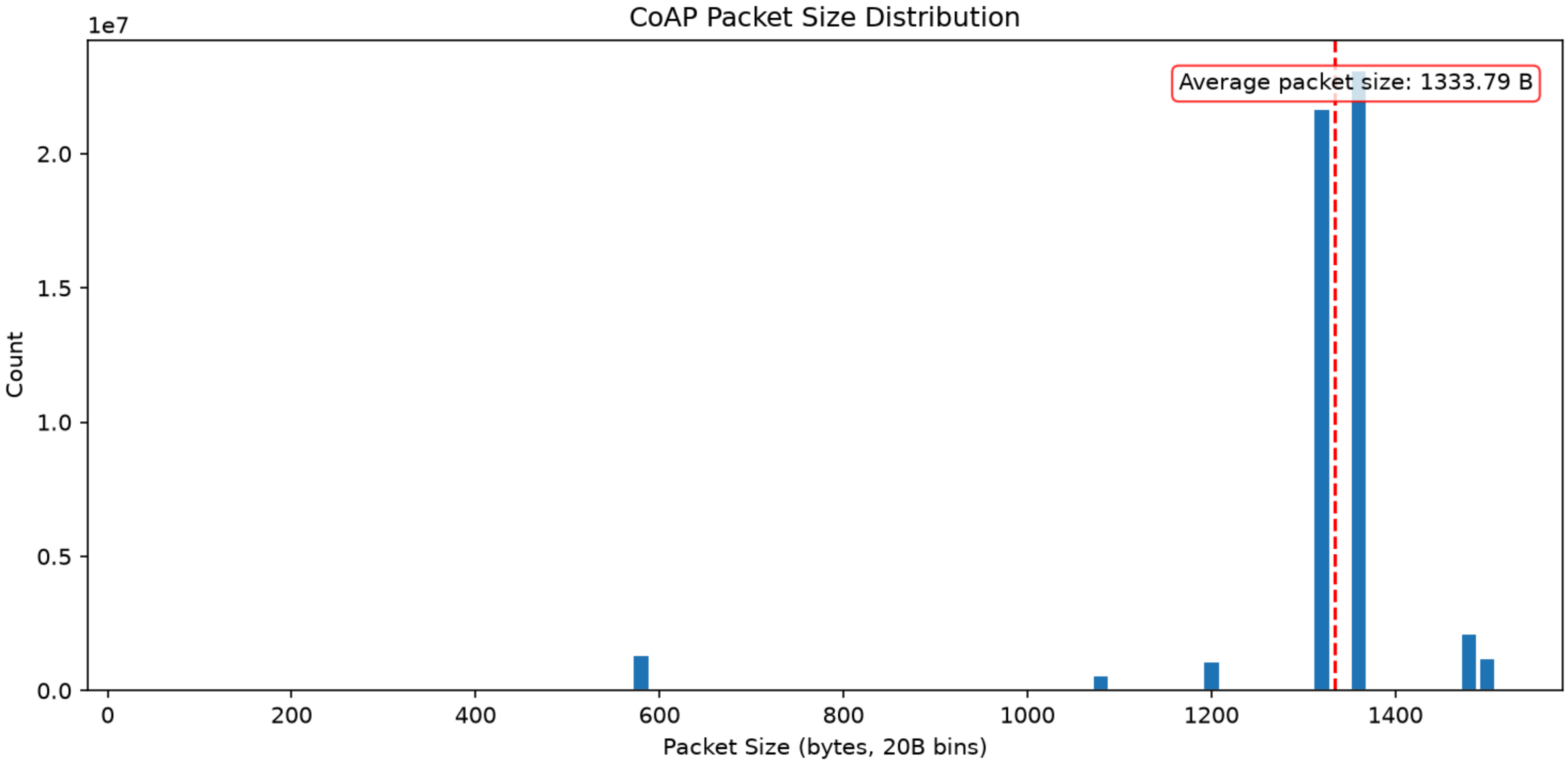

Figure 11: Packet Sizes within CoAP

CoAP specific Analysis

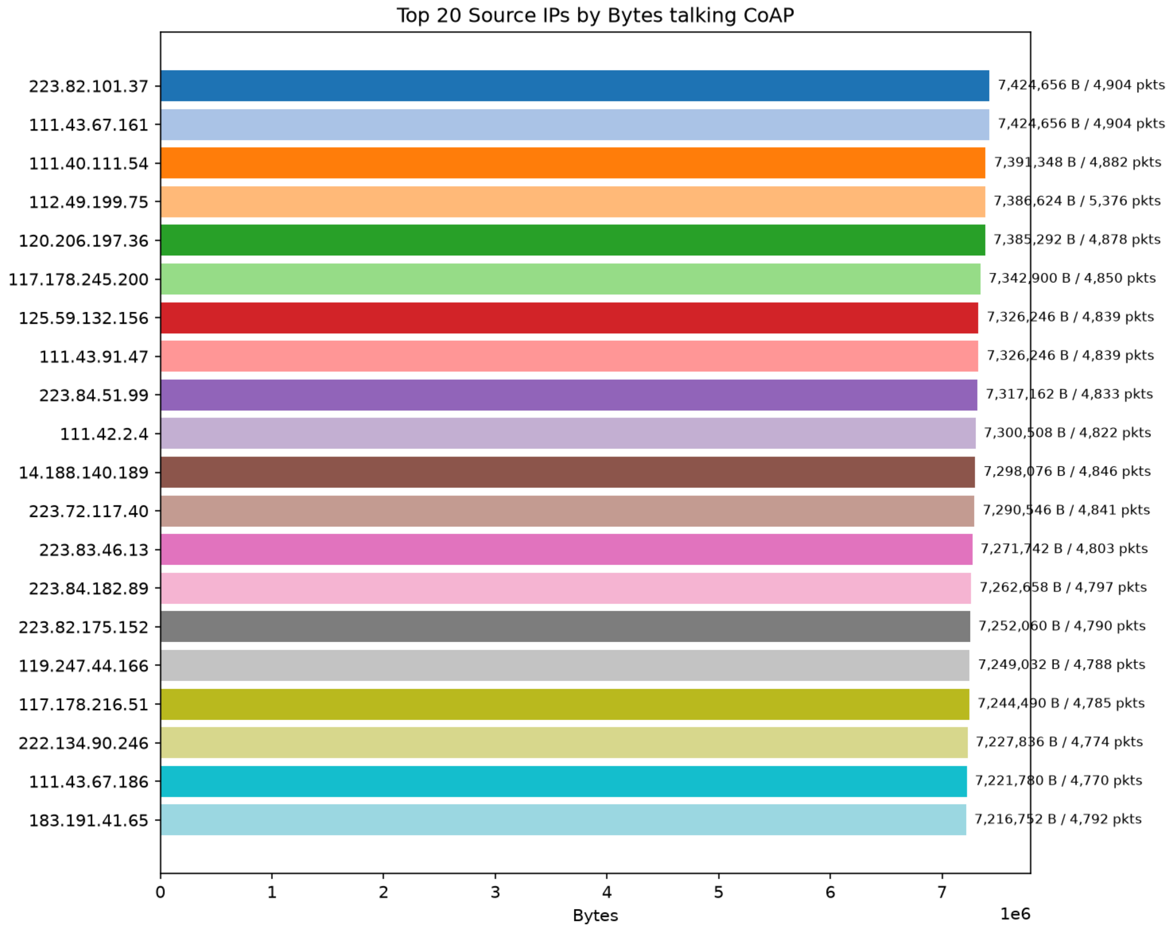

Figure 12: Top Talkers of CoAP

Outro

Answer to the Research Question

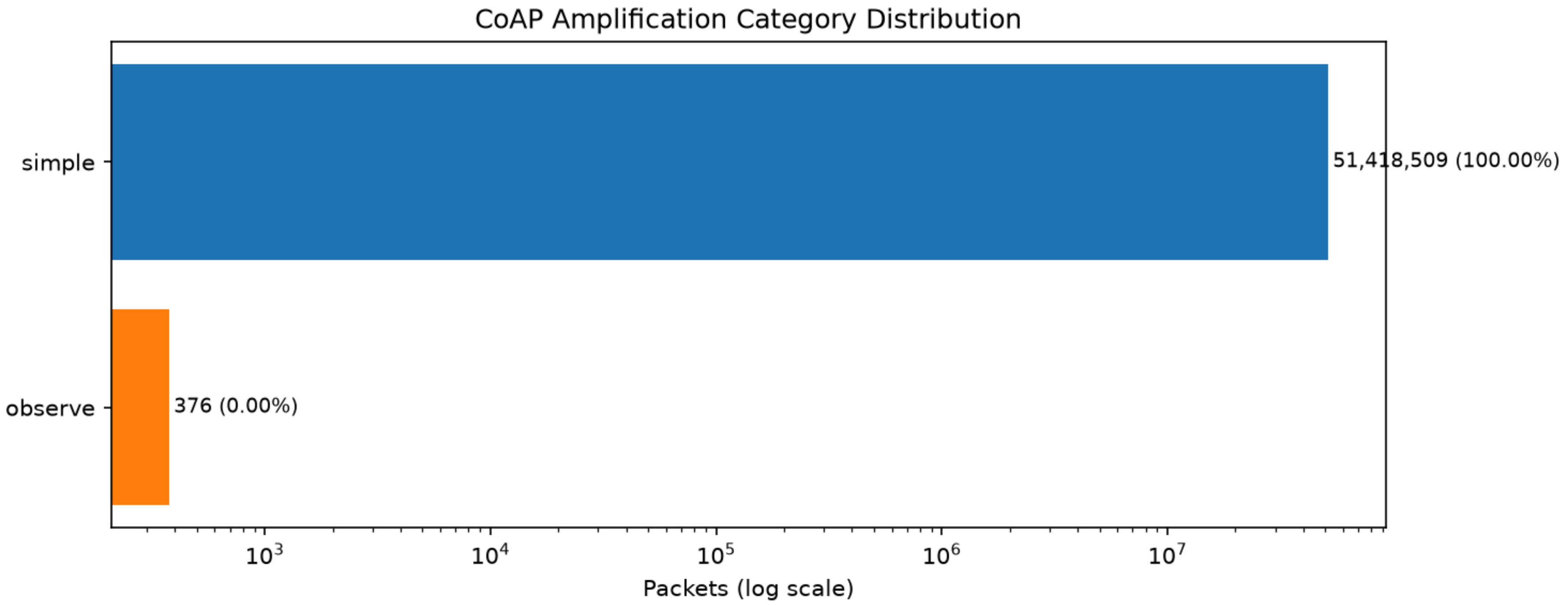

Figure 13: Amplification Attacks Categories

Answer to the Research Question

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Yes, but in the attack only Simple and Observe were used!

Other Findings

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- only around 0.7% of the DDOS attack packets consists of CoAP packets
- as expected is most of the data transported by UDP
- as expected are most of CoAP packets of method GET in a message of type ACK
- no DTLS was found

Literature

[1] J. P. Mattsson, G. Selander, and C. Amsüss, "Amplification Attacks Using the Constrained Application Protocol (CoAP)," Internet Engineering Task Force, Internet-Draft draft-irtf-t2trg-amplification-attacks-05, Jun. 2025. [Online]. Available: https://datatracker.ietf.org/doc/draft-irtf-t2trg-amplification-attacks/05/